Daily Drilling Report
SPD19B-03
3/15/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
5.0	24"	11.00	72.83	31.70	93.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
26" CP, 178.00mKB		1,913.70	606,913.73		3.60
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
357.00	491.00		134.00	15.10	8.9

Operations at Report Time
Wash down @ 467m.

Operations Summary
Directional drilling 24" hole f/ 357m to 491m. Decided to POOH due to pressure shoot up and unable to continue drilling. Sweep hole. POOH Kick off BHA f/ 491m to 27 and R/B same.

Operations next 24hrs
Cont' POOH Kick off BHA to surface. RIH Rotary BHA#3. Resume drilling.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	3/31/2014
BOP Test		
Chemical Spill Drill		
Fire Drill	3/9/2014	3/11/2014
H2S Drill		
Heli Crash Drill	2/20/2014	3/14/2014
Medical Drill		
MOB Drill	3/1/2014	3/23/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 44

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	20:00	20....	24"	D	03	P	Directional drilling 24" hole in slide & rotation mode f/ 357m to 491m w/ 800-1000 gpm / 1750-2650 psi / 5-17 kips/ 10 rpm & 3 - 11 Kft-lbs. Rotation Intervals / Slide Intervals 357m - 376m / 376m - 390m 390m - 436m / 436m - 450m 450m - 458m / 458m - 470m 470m - 484m / 484m - 489m 489m - 491m >>> Sweeping Sequence: ** Pump 20 bbl GG at middle of each jt. ** Pump 20 bbl PHG at the end of each jt. ** Pump 20 bbl GG + 20 bbl PHG at the end of stand and circulate out. ** Take Survey before connection. ** Spot 30 bbl PHG pill before connection.
20:00	21:30	1.50	24"	D	07	P	Due to pressure shoot up and unable to continue drilling decided to POOH. Stop drilling & sweep hole by pumping 40 bbls GG + Circulate for 5 min + 40 bbls PHG and circulate out w/ 990 gpm/ 2550 psi/ 10 rpm/ 2-8 Kft-lbs. Spot 100 bbl PHG pill.
21:30	00:00	2.50	24"	D	04	P	POOH Kick off BHA#2 f/ 491m to 27m. R/B BHA and retrieve MWD probe. L/D UBHO Sub.

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00		24"				Mar 16, 2014 00:00 - 00:30 Cont' POOH kick off BHA#2 f/ 27m to surface. Check the bit (1-1-WT-G-E-I-NO-BHA). *Motor Bearing Gap: 11mm. 00:30 - 03:00 M/U Rotary BHA#3 consists of new 24" Insert Bit (Type: ETR14J, IADC: 435,SN: 1250452, Nozzles: 3 x 24 + 1 x 20) + 9 1/8" Shock Sub double box (SN: COR2868AD) + 9 1/2" Float Sub (Float valve inside) + new 23" String Stabilizer (SN: COR2877AD) + 9 1/2" Restrictor Sub (With 32/32" restrictor inside) + 9 1/2" NMDC + 9 1/2" TM Sub (With MWD and Gamma probes inside) and RIH to 21m. 03:00 - 03:30 Connect TDS and perform MWD shallow test w/ 600 gpm & 570 psi, ok. 03:30 - 05:00 Cont' RIH rest of BHA consists of 9 1/2" NMDC + 17 1/2" COR Stabilizer (COR3130AD) + 1 Std 9 1/2" DC + X/O + 5 Jts 8 1/4" DC + 8" used Cougar drilling jar (COR2985AD) + 2 Jts 8 1/4" DC + X/O + 13 Jts 5 7/8" HWDP to 255m. 05:00 - 06:00 RIH Rotary BHA#3 on 5 7/8" DP f/ 255m to 467m.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
491.00	Sea Water	8.60				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
2.00	8.	330	18.0	1,014.0	22.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
F-GVEF	Chopper	3/15/2014 15:48	3/15/2014 15:52	Pax in: 0 & Pax out: 2
Morshed	Supply Boat	3/15/2014 02:10		Tranfr FO-50Mt to DP3 & DD6. Deliv cargo, Dep at 1800hrs for Lavan for further delivery of FO.
Borkan	Supply Boat	3/14/2014 23:15		Deliv 28 MT CMT. Departed to Kish at 1815hrs.ETA@ 0900 Hrs/16.03.14.
DP3	Supply Boat	2/23/2014 23:30		Received FW-150MT from Borkan. Standby Anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Barite Bulk	0.0	221.0	221.0
Caustic Soda	1.0	427.0	266.0
Lime	1.0	150.0	67.0
Guar Gam MP	3.0	1,421.0	1,199.0
Soda Ash	1.0	151.0	53.0
Bentonite Bulk	9.0	105.0	81.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Cement G	0.0	28.0	183.0
Diesel Fuel	11.0	50.0	201.0
Potable Water	18.0	24.0	85.0
Drill Water	95.0	0.0	455.0
Barite Bulk	0.0	0.0	209.0
Bentonite BB	0.0	0.0	45.0
Bentonite Bulk	9.0	0.0	73.0
Barite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
2	24	Varel	ETR1GJMRS	1239954	Mill Tooth	115	20/20/20/20
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.22	27.65	173,587.20	183.00	491.00	308.00	11.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
600	1,000	1,000.0	2,650.0	5	17	69	126
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
37.28	18.37	66.44	33.56	230	221	195	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
357.00	376.00	1.72	8,000.0	800	10	1,850.0	15	215
Depth Start (mKB)	**Depth End (mKB)**	**Drilling Time (hrs)**	**Drilling Torque**	**Flow Rate (gpm)**	**RPM (rpm)**	**SPP (psi)**	**WOB (1000lbf)**	**Drill Str Wt (1000lbf)**
376.00	390.00	0.53		800		1,850.0	15	215

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Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
390.00	436.00	8.55	8,000.0	900	10	2,250.0	17	216
436.00	450.00	0.70		950		2,350.0	17	216
450.00	458.00	0.47	9,000.0	1,000	10	2,650.0	15	220
458.00	470.00	0.63		950		2,350.0	10	220
470.00	484.00	1.73	10,000.0	1,000	10	2,650.0	15	221
484.00	489.00	0.48		990		2,540.0	5	221
489.00	491.00	0.29	4,000.0	990	10	2,540.0	10	221

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
13 Jts 5 7/8" HWDP	5 7/8	4.250	120.96	259.51
X/O	7	2.810	0.79	138.55
2 Jts 8 1/4" DC	8 1/4	2.810	18.75	137.76
8" Drilling Cougar Jar	8	3.500	5.84	119.01
5 Jts 8 1/4" DC	8 1/4	2.810	47.24	113.17
X/O	9 5/8	3.000	1.22	65.93
3 Jts 9 1/2" DC	9 1/2	3.000	28.23	64.71
9 1/2" NMDC	9 1/2	3.100	9.44	36.48
9 1/2" UBHO Sub	9 1/2	3.000	0.83	27.04
9 1/2" Float sub (F.valve inside)	9 1/2		0.77	26.21
9 1/2" Pulser Sub	9 1/2		2.15	25.44
9 1/2" NMDC	9 1/2	4.000	9.46	23.29
9 1/2" Restrictor Sub with restrictor	9 1/2	4.250	0.92	13.83
9 1/2" PNMDC	9 1/2		3.05	12.91
11 1/4" Motor (AKO: 1.83° , Sleeve: 23 1/4")	11 1/4		9.25	9.86
24" Mill Tooth Bit	24		0.61	0.61

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
343.44	228.00	10.90	342.18	-13.20	-13.16	18.63	0.95	0.93
372.64	227.50	12.50	370.77	-17.19	-17.54	24.55	1.65	1.64
401.80	228.40	12.70	399.23	-21.45	-22.26	30.91	0.29	0.21
431.00	229.00	14.40	427.61	-25.96	-27.41	37.75	1.75	1.75
460.00	230.00	15.60	455.63	-30.83	-33.11	45.25	1.27	1.24

General Remarks

Comment

>>> Requirements: (23" Stabilizer) + (Shock Sub) + (200 MT DW).

>>> Sent:Nil.

>>> Received:
*** Morshed f/ Sharjah: (Aban Equipment).

*** Borkan f/ Kish: (28 MT CMT).